

genaut — frontier report

source CSV : genaut/logs/1statelap3acc/default.csv
corpus rows: 7

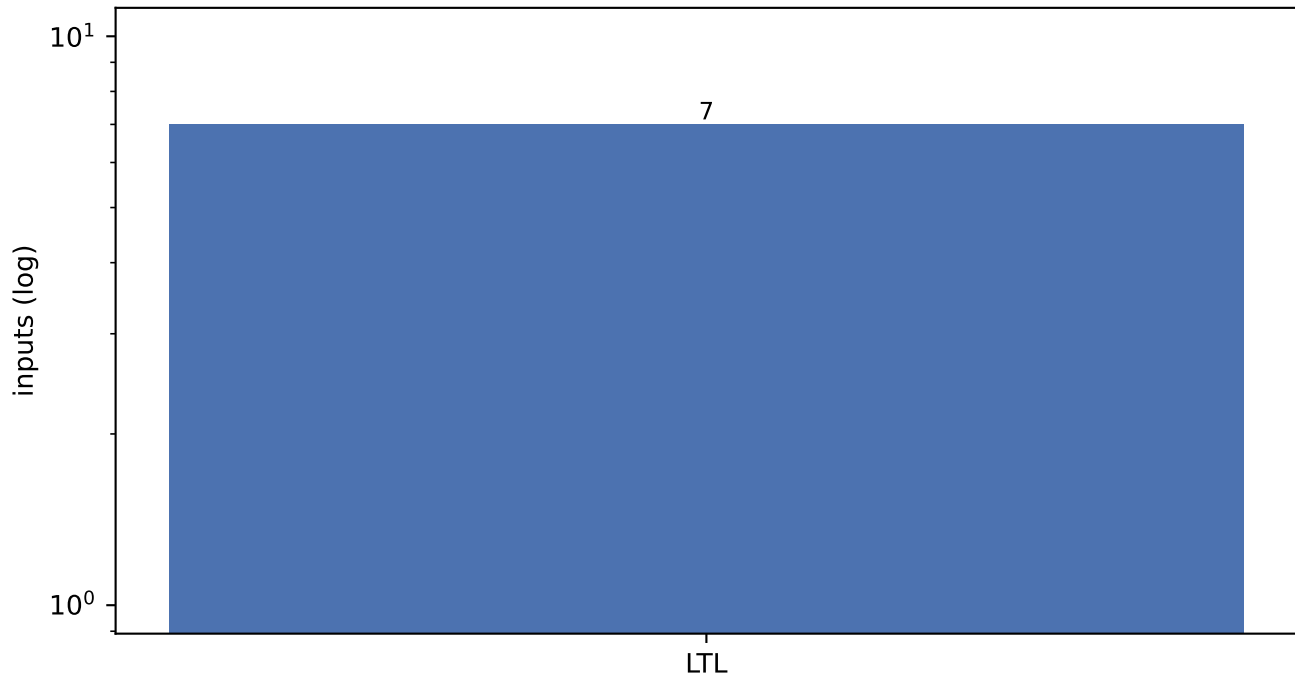
sections:

- Ventilation – LTL / not-LTL / ambiguous
- Formula collapse – structural (md5) concentration
- Size frontier – flattened formula size
- Routes to `true` – which technique built the universal 1
- Build time & timeouts
- Verification – spot-oracle validation

Ventilation — LTL / not-LTL / ambiguous

```
corpus inputs: 7
  LTL-definable (formula built) :    7  (100.0%)
    not-LTL (decided)           :    0  (  0.0%)
    ambiguous (timeout, unknown) :    0  (  0.0%)
decided (LTL or not-LTL): 7  (100.0%)
```

Answer frontier (by result)



Formula collapse — structural (md5) concentration

LTL answers: 7 (0 with a gated/unprintable string)

distinct structures (md5): 5

80% of answers covered by the top 4 structure(s)

90% of answers covered by the top 5 structure(s)

top 15 structures:

2 (28.6%) 0

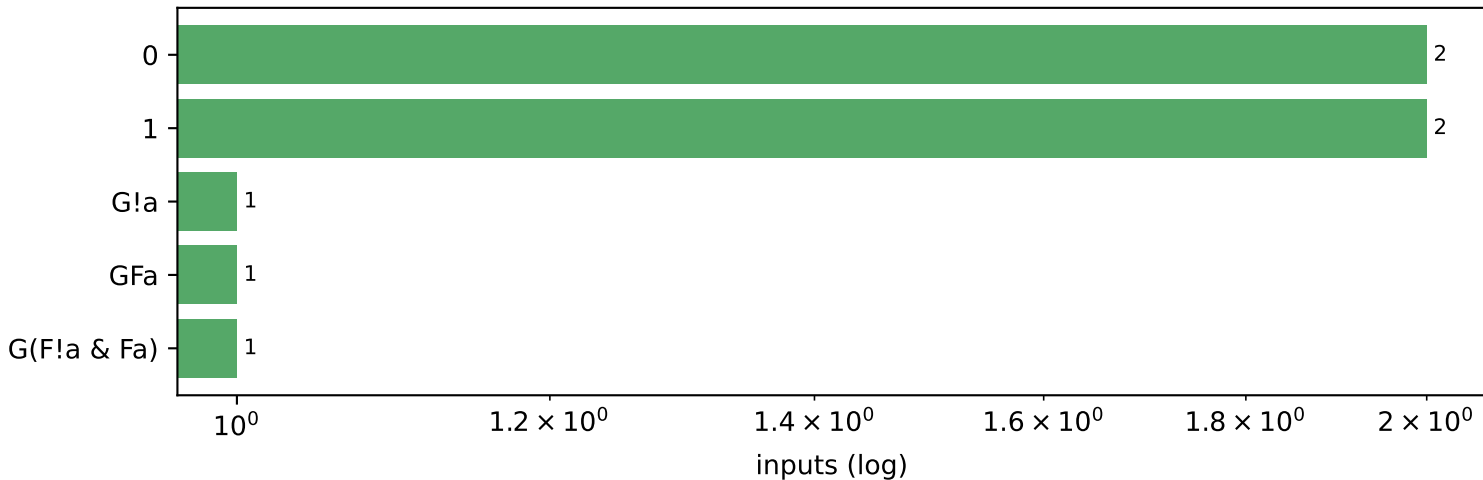
2 (28.6%) 1

1 (14.3%) $G!a$

1 (14.3%) GFa

1 (14.3%) $G(F!a \ \& \ Fa)$

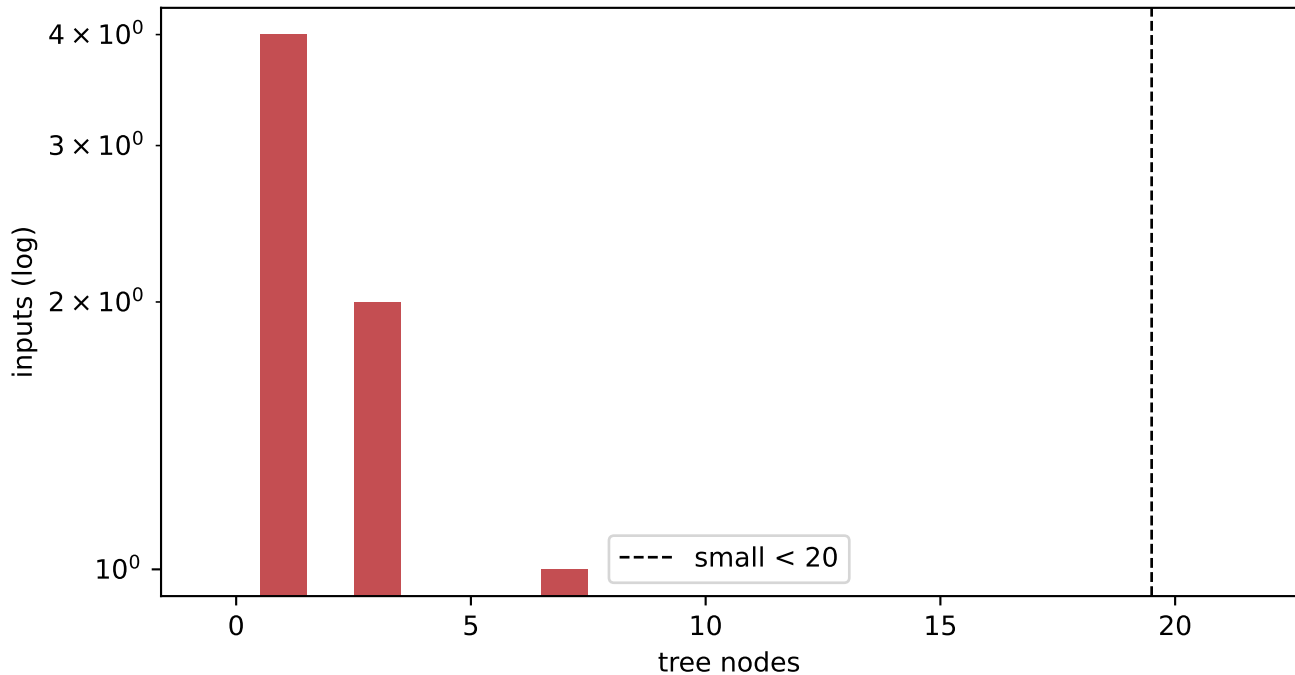
Top 5 reconstructed structures



Size frontier — flattened formula size

```
LTL answers with a size: 7
trivial (tree<=1, i.e. true/false): 4 ( 57.1%)
small   (tree<20):                    7 (100.0%)
larger  (tree>=20):                   0 (  0.0%)
tree nodes: median 1  mean 2.4  max 7
```

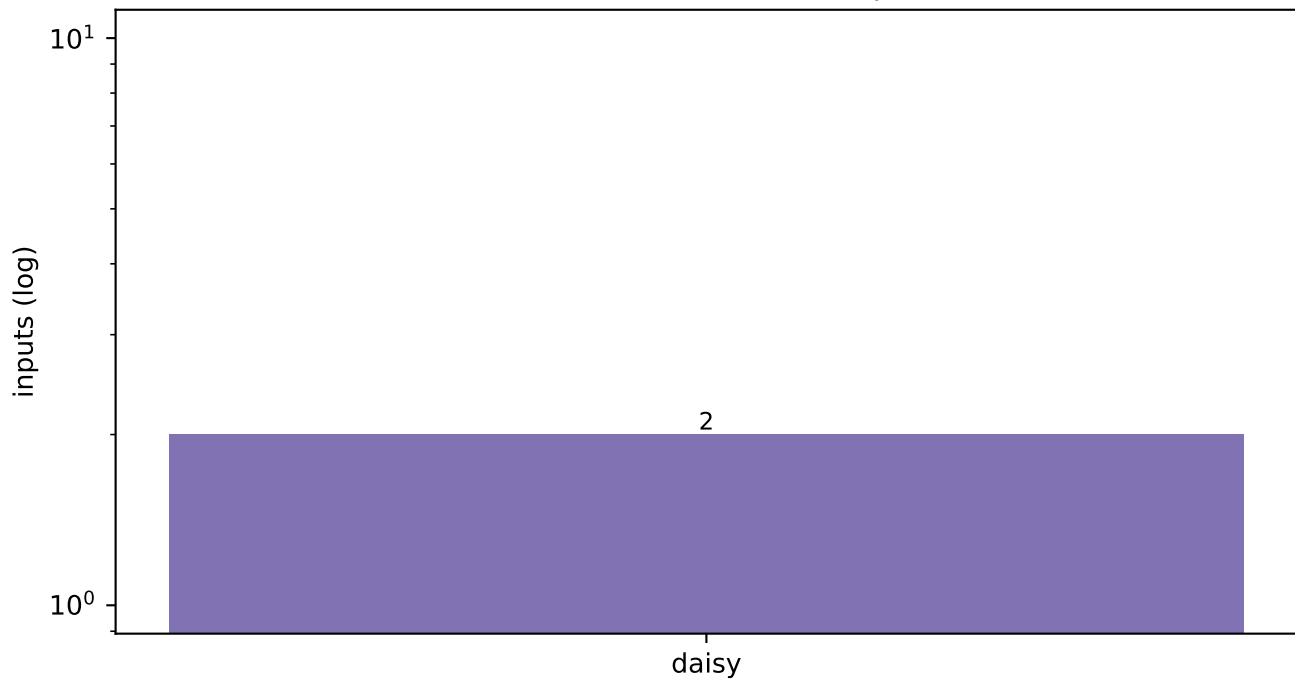
Flattened size (tree nodes); clipped at 21 (0 larger pooled at right)



Routes to `true` — which technique built the universal 1

```
universal answers (formula == '1'): 2  
  2 (100.0%) daisy  
build_s over true rows: total 0.1s  mean 0.070s  max 0.071s
```

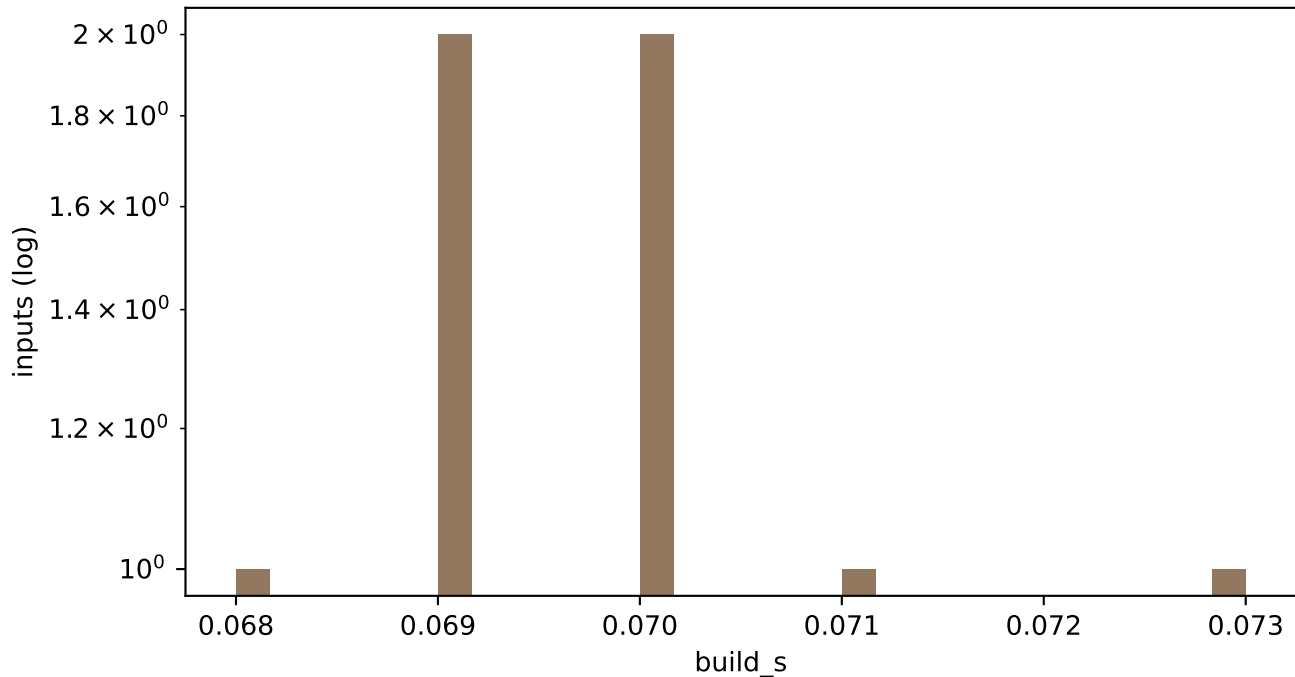

Routes to `true` (technique)



Build time & timeouts

```
total build: 0.5s    mean 0.070s    max 0.073s
  LTL          n=    7    mean 0.070s    max 0.073s
timeouts: 0
```

Build-time distribution



Verification — spot-oracle validation

```
validation over 7 LTL answers:  
  True          7 (100.0%)  
NOT-equivalent (FALSE): 0    CLEAN
```